

Decision stability under reseeding, at realistic misfit

How often does the same data give a different answer?

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2026-08-01

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Scope

Error rates averaged over datasets can look fine while individual decisions remain unstable. This study holds the **dataset fixed** and varies only the bootstrap seed, isolating Monte-Carlo noise from sampling noise: if an analyst re-runs `RMitemInfitCutoff()` on the same data with a different seed, how often does the flag change?

An earlier version of this exists in `infit_fwer_iterations.qmd` (Study B), but it has two limitations that this run addresses:

1. It used **dichotomous** items at $\rho = 0.15$ — misfit roughly 60 percent stronger than anything observed in real phq9 data, which we now know saturates detection.
2. It covered only n in $\{250, 500\}$ and $B \leq 1000$.

This matters because the current recommendation of $B \geq 1000$ for FWER rests entirely on that single measurement — stability, not calibration, is what drives it. Calibration is already nominal from $B \approx 400$.

Prediction being tested

Instability should be **worse** at realistic misfit, because when the signal is weaker more items sit near the decision boundary, and a boundary-adjacent item is exactly the one a different seed can push either way. Three levels are included so this is a dose-response test rather than a two-point comparison: $\rho = 0.15$ (the unrealistically strong condition used in the earlier study), $\rho = 0.50$ (intermediate), and $\rho = 0.85$ (calibrated to real phq9 misfit, injected `infit` ~ 1.25 against $q3 = 1.23$ and $q8 = 1.26$).

Design

- Same two arms as `infit_width_iterations_poly_weak.qmd`: **sim15** (15 stylised polytomous items) and **phq9** (9 items, CML parameters from the packaged real responses)
- ρ in $\{0.15, 0.50, 0.85\}$ for the misfit conditions; $k = 0$ has no ρ
- Misfitting items: the maximum per arm (3 for sim15, 2 for phq9), i.e. the strongest cascade, plus a $k = 0$ null baseline
- n in $\{250, 1000\}$
- **3 fixed datasets** per condition, each analysed with **25 bootstrap seeds** (seeds resolve instability better than datasets do — an item flagged with probability 0.1 is caught as non-unanimous 93 percent of the time with 25 seeds, only 79 percent with 15)
- B in $\{100, 400, 1000, 2000\}$; widths $\{.90, .95, .99, .995, .999\}$ read off the same draws
- FWER and BH p-values recorded alongside

Outcomes: the share of items whose flag is not unanimous across seeds, the probability that two randomly chosen seeds disagree, and the spread of the adjusted p-value and interval bounds.

```
suppressMessages({
  pkgload::load_all("../", quiet = TRUE)
  library(ggdist)
  library(MASS)
  library(parallel)
```

```

library(dplyr)
library(tidyr)
library(ggplot2)
})

WIDTHS <- c(0.90, 0.95, 0.99, 0.995, 0.999)
NS <- c(250, 1000)
BS <- c(100, 400, 1000, 2000)
RHOS <- c(0.15, 0.50, 0.85)
N_DATA <- 6 # raised from 3 on 2026-07-31; datasets 4-6 resume
N_SEEDS <- 25 # bootstrap seeds per dataset
NCORES <- 10
ALPHA <- 0.05
CHUNK <- 30 # frequent checkpoints: this run may be stopped mid-way
RES <- "infit_decision_stability.rds"

```

```

run_resumable <- function(res_file, grid, keys, fn,
                          ncores = NCORES, chunk_size = CHUNK,
                          extra = list()) {
  key_of <- function(g) {
    if (nrow(g) == 0L) return(character(0))
    do.call(paste, c(unname(as.list(g[keys])), sep = "|"))
  }
  acc <- if (file.exists(res_file)) readRDS(res_file) else
    c(list(res = list(), grid = grid[0, , drop = FALSE], elapsed_min = 0), extra)
  todo <- which(!(key_of(grid) %in% key_of(acc$grid)))
  if (length(todo) == 0L) {
    message("cached: ", nrow(acc$grid), " cells, nothing to run"); return(acc)
  }
  message("running ", length(todo), " new cells (", nrow(acc$grid), " cached)")
  for (s in seq(1, length(todo), by = chunk_size)) {
    idx <- todo[s:min(s + chunk_size - 1L, length(todo))]
    t0 <- Sys.time()
    out <- mclapply(idx, function(i) fn(grid[i, , drop = FALSE]), mc.cores = ncores)
    acc$res <- c(acc$res, out)
    acc$grid <- rbind(acc$grid, grid[idx, , drop = FALSE])
    acc$elapsed_min <- acc$elapsed_min + as.numeric(Sys.time() - t0, "mins")
    acc <- utils::modifyList(acc, extra)
    saveRDS(acc, res_file)
    message(" checkpoint ", nrow(acc$grid), " / ", nrow(grid),
            " (", round(acc$elapsed_min, 1), " min)")
  }
  acc
}

```

Arms (identical to the poly_weak study)

```

S15_LOCS <- seq(-2, 2, length.out = 15)
S15_THR <- lapply(S15_LOCS, function(L) L + seq(-1, 1, length.out = 3))
S15_MIS <- vapply(c(0, -1, -2), function(tg) which.min(abs(S15_LOCS - tg)),
                 integer(1))

load("../data/phq9.rda")
PH_OBS <- as.matrix(phq9[, paste0("q", 1:9)])
PH_OBS <- PH_OBS[complete.cases(PH_OBS), ]
PH_THR <- .fit_cml_thresholds(PH_OBS)
PH_LOCS <- vapply(PH_THR, mean, numeric(1))
.g <- seq(-6, 6, length.out = 81)
PH_SD <- .estimate_prior_sd(
  .grid_loglik(PH_OBS, .logp_tables(PH_THR, .g), .g), .g, 0)
PH_MIS <- c(5L, 8L)

```

```

ARMS <- list(
  sim15 = list(thr = S15_THR, sd = 1.5, mis = S15_MIS, locs = S15_LOCS,
    nm = paste0("V", 1:15), kmax = 3L),
  phq9 = list(thr = PH_THR, sd = PH_SD, mis = PH_MIS, locs = PH_LOCS,
    nm = paste0("q", 1:9), kmax = 2L)
)

sim_arm <- function(arm, n, k, rho, seed) {
  a <- ARMS[[arm]]
  set.seed(seed)
  S <- matrix(c(a$sd^2, rho * a$sd^2, rho * a$sd^2, a$sd^2), 2, 2)
  th <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = S)
  ni <- length(a$thr); mis <- a$mis[seq_len(k)]
  d <- matrix(0L, n, ni); fit_items <- setdiff(seq_len(ni), mis)
  if (length(fit_items)) d[, fit_items] <- sim_partial_score(a$thr[fit_items], th[, 1])
  if (length(mis)) d[, mis] <- sim_partial_score(a$thr[mis], th[, 2])
  d <- as.data.frame(d); names(d) <- a$nm; d
}

bounds_at <- function(results, width, items) {
  vapply(items, function(it) {
    v <- results$InfitMSQ[results$Item == it]
    as.numeric(ggdist::hdc1(v, .width = width))[1:2]
  }, numeric(2))
}

flag_code <- function(msq, lo, hi) ifelse(msq > hi, 2L, ifelse(msq < lo, 1L, 0L))

```

Run

Each cell is one (dataset, seed) pair: the dataset is regenerated from a seed that depends only on (arm, k, n, rho, dataset), so it is **identical across all bootstrap seeds**, while the bootstrap seed varies.

```

one_cell <- function(g) {
  out <- try({
    arm <- as.character(g$arm); k <- g$k; n <- g$n; rho <- g$rho
    dat <- g$dataset; sd_seed <- g$seed
    d <- sim_arm(arm, n, k, rho,
      seed = 5e8 + 1e7 * match(arm, names(ARMS)) + 1e6 * k +
        1e4 * round(rho * 100) + 1e3 * n + dat)
    items <- names(d)
    obs <- RMitemInfit(d, output = "dataframe")
    msq <- obs$Infit_MSQ[match(items, obs$Item)]
    hd <- list(); pv <- list()
    for (j in seq_along(BS)) {
      co <- try(RMitemInfitCutoff(d, iterations = BS[j], parallel = FALSE,
        seed = sd_seed * 1000 + j, verbose = FALSE),
        silent = TRUE)
      if (inherits(co, "try-error")) next
      hd[[length(hd) + 1L]] <- do.call(rbind, lapply(WIDTHS, function(w) {
        b <- bounds_at(co$results, w, items)
        data.frame(B = BS[j], w = w, item = seq_along(items),
          code = flag_code(msq, b[1, ], b[2, ]),
          lo = b[1, ], hi = b[2, ])
      })))
      fw <- try(suppressWarnings(
        RMitemInfit(d, cutoff = co, p_value = TRUE, correction = "fwer",
          alpha = ALPHA, output = "dataframe")), silent = TRUE)
      bh <- try(suppressWarnings(
        RMitemInfit(d, cutoff = co, p_value = TRUE, correction = "fdr_bh",
          alpha = ALPHA, output = "dataframe")), silent = TRUE)
      if (!inherits(fw, "try-error") && !inherits(bh, "try-error")) {
        i1 <- match(items, fw$Item); i2 <- match(items, bh$Item)
        pv[[length(pv) + 1L]] <- data.frame(

```

```

      B = BS[j], item = seq_along(items),
      padj_fwer = fw$padj_infit[i1], padj_bh = bh$padj_infit[i2])
    }
  }
  list(hdci = do.call(rbind, hd), pval = do.call(rbind, pv))
}, silent = TRUE)
if (inherits(out, "try-error")) NULL else out
}

blocks <- list()
for (a in names(ARMS)) {
  # null baseline: rho is inert with no misfitting items
  blocks[[length(blocks) + 1L]] <- expand.grid(
    arm = a, k = 0L, rho = 0, n = NS, dataset = 1:N_DATA, seed = 1:N_SEEDS,
    stringsAsFactors = FALSE)
  blocks[[length(blocks) + 1L]] <- expand.grid(
    arm = a, k = ARMS[[a]]$kmax, rho = RHOS, n = NS,
    dataset = 1:N_DATA, seed = 1:N_SEEDS, stringsAsFactors = FALSE)
}
grid <- do.call(rbind, blocks)[, c("arm", "k", "rho", "n", "dataset", "seed")]

# Explicit priority order, so a run stopped early still answers the main
# question. Blocks cost roughly the same (~2.3 h each on 10 cores):
# 1. rho = 0.85 -- the calibrated, realistic condition (the point of the run)
# 2. k = 0      -- instability under a true model, the reference baseline
# 3. rho = 0.15 -- the old unrealistically strong condition, for contrast
# 4. rho = 0.50 -- intermediate; fills in the dose-response, least critical
prio <- with(grid, ifelse(k == 0L, 2L,
  ifelse(rho == 0.85, 1L,
    ifelse(rho == 0.15, 3L, 4L))))
grid <- grid[order(prio, grid$n), ]
rownames(grid) <- NULL
cat("cells to run:", nrow(grid), "\n")

```

```
cells to run: 2400
```

```
print(table(priority = sort(prio)))
```

```

priority
 1  2  3  4
600 600 600 600

```

```

s <- run_resumable(RES, grid, c("arm", "k", "rho", "n", "dataset", "seed"),
  one_cell,
  extra = list(arms = lapply(ARMS, function(a)
    a[c("mis", "locs", "sd", "nm", "kmax")))))

bind_part <- function(part) {
  do.call(rbind, lapply(seq_len(nrow(s$grid)), function(i) {
    r <- s$res[[i]][[part]]
    if (is.null(r)) return(NULL)
    cbind(s$grid[i, c("arm", "k", "rho", "n", "dataset", "seed")], r)
  })))
}
H <- bind_part("hdci") |> mutate(flag = as.integer(code > 0),
  w = sprintf("%.3f", w),
  rho = sprintf("%.2f", rho))
P <- bind_part("pval") |> mutate(rho = sprintf("%.2f", rho))
cat(nrow(s$grid), "cells |", round(s$elapsed_min, 1), "min |",
  "failed:", sum(sapply(s$res, is.null)), "\n")

```

Instability of the HDCl rule

Share of items whose flag is **not unanimous** across the 25 seeds, averaged over the 5 datasets.

```
H |>
  group_by(arm, rho, k, n, w, B, dataset, item) |>
  summarise(rate = mean(flag), .groups = "drop") |>
  group_by(arm, rho, k, n, w, B) |>
  summarise(unstable = round(100 * mean(rate > 0 & rate < 1), 1), .groups = "drop") |>
  filter(w == "0.999") |>
  pivot_wider(names_from = B, values_from = unstable, names_prefix = "B=") |>
  arrange(arm, rho, n)
```

arm	rho	k	n	w	B=100	B=400	B=1000	B=2000
phq9	0.00	0	250	0.999	3.7	1.9	1.9	1.9
phq9	0.00	0	1000	0.999	7.4	0.0	0.0	0.0
phq9	0.15	2	250	0.999	37.0	35.2	24.1	22.2
phq9	0.15	2	1000	0.999	1.9	1.9	1.9	3.7
phq9	0.50	2	250	0.999	46.3	29.6	18.5	14.8
phq9	0.50	2	1000	0.999	42.6	40.7	31.5	33.3
phq9	0.85	2	250	0.999	16.7	7.4	5.6	3.7
phq9	0.85	2	1000	0.999	27.8	20.4	14.8	13.0
sim15	0.00	0	250	0.999	4.4	2.2	0.0	0.0
sim15	0.00	0	1000	0.999	6.7	2.2	0.0	1.1
sim15	0.15	3	250	0.999	51.1	36.7	25.6	25.6
sim15	0.15	3	1000	0.999	6.7	10.0	8.9	7.8
sim15	0.50	3	250	0.999	35.6	25.6	15.6	12.2
sim15	0.50	3	1000	0.999	41.1	43.3	34.4	28.9
sim15	0.85	3	250	0.999	13.3	4.4	5.6	3.3
sim15	0.85	3	1000	0.999	22.2	12.2	7.8	5.6

Instability of the FWER and BH rules

```
P |>
  mutate(fwer = as.integer(padj_fwer < ALPHA),
         bh = as.integer(padj_bh < ALPHA)) |>
  pivot_longer(c(fwer, bh), names_to = "rule", values_to = "flag") |>
  group_by(arm, rho, k, n, rule, B, dataset, item) |>
  summarise(rate = mean(flag, na.rm = TRUE), .groups = "drop") |>
  group_by(arm, rho, k, n, rule, B) |>
  summarise(unstable = round(100 * mean(rate > 0 & rate < 1), 1), .groups = "drop") |>
  pivot_wider(names_from = B, values_from = unstable, names_prefix = "B=") |>
  arrange(rule, arm, rho, n)
```

arm	rho	k	n	rule	B=100	B=400	B=1000	B=2000
phq9	0.00	0	250	bh	0.0	1.9	0.0	0.0
phq9	0.00	0	1000	bh	0.0	0.0	0.0	0.0
phq9	0.15	2	250	bh	31.5	11.1	9.3	7.4

arm	rho	k	n	rule	B=100	B=400	B=1000	B=2000
phq9	0.15	2	1000	bh	0.0	0.0	0.0	0.0
phq9	0.50	2	250	bh	46.3	27.8	24.1	18.5
phq9	0.50	2	1000	bh	7.4	5.6	5.6	5.6
phq9	0.85	2	250	bh	11.1	3.7	3.7	1.9
phq9	0.85	2	1000	bh	24.1	9.3	3.7	5.6
sim15	0.00	0	250	bh	0.0	0.0	0.0	0.0
sim15	0.00	0	1000	bh	0.0	0.0	0.0	0.0
sim15	0.15	3	250	bh	42.2	21.1	11.1	10.0
sim15	0.15	3	1000	bh	0.0	0.0	0.0	0.0
sim15	0.50	3	250	bh	38.9	18.9	14.4	10.0
sim15	0.50	3	1000	bh	24.4	8.9	7.8	4.4
sim15	0.85	3	250	bh	0.0	4.4	3.3	2.2
sim15	0.85	3	1000	bh	21.1	8.9	6.7	4.4
phq9	0.00	0	250	fwer	1.9	0.0	0.0	0.0
phq9	0.00	0	1000	fwer	1.9	0.0	0.0	0.0
phq9	0.15	2	250	fwer	37.0	25.9	22.2	16.7
phq9	0.15	2	1000	fwer	0.0	0.0	0.0	0.0
phq9	0.50	2	250	fwer	25.9	9.3	9.3	7.4
phq9	0.50	2	1000	fwer	13.0	11.1	11.1	5.6
phq9	0.85	2	250	fwer	9.3	3.7	3.7	1.9
phq9	0.85	2	1000	fwer	13.0	7.4	5.6	3.7
sim15	0.00	0	250	fwer	1.1	0.0	0.0	0.0
sim15	0.00	0	1000	fwer	1.1	0.0	0.0	0.0
sim15	0.15	3	250	fwer	36.7	15.6	11.1	10.0
sim15	0.15	3	1000	fwer	0.0	0.0	0.0	0.0
sim15	0.50	3	250	fwer	15.6	5.6	2.2	2.2
sim15	0.50	3	1000	fwer	31.1	16.7	8.9	5.6
sim15	0.85	3	250	fwer	3.3	2.2	2.2	1.1
sim15	0.85	3	1000	fwer	6.7	2.2	2.2	1.1

Probability that two analysts disagree

For an item flagged with probability p across seeds, two independently seeded runs disagree with probability $2p(1-p)$. Averaged over items, this is the chance that two analysts running the same data reach a different conclusion about at least one item.

```
P |>
  mutate(fwer = as.integer(padj_fwer < ALPHA)) |>
  group_by(arm, rho, k, n, B, dataset, item) |>
  summarise(p = mean(fwer, na.rm = TRUE), .groups = "drop") |>
  mutate(disagree = 2 * p * (1 - p)) |>
  group_by(arm, rho, k, n, B) |>
  summarise(any_item = round(100 * (1 - prod(1 - disagree)^(1 / n_distinct(dataset))), 1),
            per_item = round(100 * mean(disagree), 2), .groups = "drop") |>
  pivot_wider(names_from = B, values_from = c(per_item, any_item)) |>
  arrange(arm, rho, n)
```

arm	rhok	n	per_item_100	per_item_400	per_item_1000	per_item_2000	any_item_100	any_item_400	any_item_1000	any_item_2000
phq9	0.000	250	0.68	0.00	0.00	0.00	7.3	0.0	0.0	0.0
phq9	0.000	1000	0.14	0.00	0.00	0.00	1.3	0.0	0.0	0.0
phq9	0.152	250	12.34	7.85	6.04	3.93	75.9	59.9	49.7	34.7
phq9	0.152	1000	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.0
phq9	0.502	250	7.40	3.47	2.30	1.46	56.4	33.2	22.4	14.3
phq9	0.502	1000	5.43	2.71	2.41	1.39	47.9	25.3	23.1	13.4
phq9	0.852	250	2.13	0.64	0.55	0.27	20.8	6.3	5.2	2.6
phq9	0.852	1000	4.27	2.37	1.30	1.00	38.9	23.8	13.8	11.0
sim15	0.000	250	0.09	0.00	0.00	0.00	1.3	0.0	0.0	0.0
sim15	0.000	1000	0.23	0.00	0.00	0.00	3.9	0.0	0.0	0.0
sim15	0.153	250	10.30	4.98	2.74	2.15	85.9	61.9	39.5	31.8
sim15	0.153	1000	0.00	0.00	0.00	0.00	0.0	0.0	0.0	0.0
sim15	0.503	250	3.91	1.14	0.63	0.52	51.3	18.5	11.9	8.7
sim15	0.503	1000	8.56	3.56	2.82	1.12	80.2	46.9	41.8	18.2
sim15	0.853	250	0.97	0.77	0.68	0.41	16.8	13.8	12.1	7.3
sim15	0.853	1000	2.03	0.87	0.79	0.55	32.1	15.4	14.3	10.9

Spread of the adjusted p-value

```
P |>
  group_by(arm, rho, k, n, B, dataset, item) |>
  summarise(sd_p = sd(padj_fwer, na.rm = TRUE), .groups = "drop") |>
  group_by(arm, rho, k, n, B) |>
  summarise(median_sd = round(median(sd_p, na.rm = TRUE), 4), .groups = "drop") |>
  pivot_wider(names_from = B, values_from = median_sd, names_prefix = "B=") |>
  arrange(arm, rho, n)
```

arm	rho	k	n	B=100	B=400	B=1000	B=2000
phq9	0.00	0	250	0.0105	0.0065	0.0034	0.0029
phq9	0.00	0	1000	0.0240	0.0126	0.0090	0.0049
phq9	0.15	2	250	0.0197	0.0093	0.0061	0.0045
phq9	0.15	2	1000	0.0000	0.0000	0.0000	0.0000
phq9	0.50	2	250	0.0487	0.0248	0.0174	0.0112
phq9	0.50	2	1000	0.0030	0.0019	0.0010	0.0007
phq9	0.85	2	250	0.0395	0.0225	0.0166	0.0090
phq9	0.85	2	1000	0.0677	0.0312	0.0229	0.0164
sim15	0.00	0	250	0.0081	0.0050	0.0033	0.0024
sim15	0.00	0	1000	0.0073	0.0042	0.0024	0.0018
sim15	0.15	3	250	0.0354	0.0207	0.0128	0.0093
sim15	0.15	3	1000	0.0000	0.0000	0.0000	0.0000
sim15	0.50	3	250	0.0563	0.0326	0.0204	0.0159
sim15	0.50	3	1000	0.0098	0.0051	0.0032	0.0023
sim15	0.85	3	250	0.0066	0.0032	0.0021	0.0015

arm	rho	k	n	B=100	B=400	B=1000	B=2000
sim15	0.85	3	1000	0.0261	0.0123	0.0102	0.0059

Spread of the HDCI bounds

```
H |>
  filter(w == "0.999") |>
  group_by(arm, rho, k, n, B, dataset, item) |>
  summarise(sd_hi = sd(hi), .groups = "drop") |>
  group_by(arm, rho, k, n, B) |>
  summarise(median_sd_upper = round(median(sd_hi), 4), .groups = "drop") |>
  pivot_wider(names_from = B, values_from = median_sd_upper, names_prefix = "B=") |>
  arrange(arm, rho, n)
```

arm	rho	k	n	B=100	B=400	B=1000	B=2000
phq9	0.00	0	250	0.0400	0.0365	0.0275	0.0226
phq9	0.00	0	1000	0.0179	0.0159	0.0118	0.0104
phq9	0.15	2	250	0.0351	0.0314	0.0226	0.0203
phq9	0.15	2	1000	0.0164	0.0149	0.0110	0.0094
phq9	0.50	2	250	0.0387	0.0337	0.0251	0.0226
phq9	0.50	2	1000	0.0169	0.0157	0.0110	0.0105
phq9	0.85	2	250	0.0409	0.0345	0.0259	0.0240
phq9	0.85	2	1000	0.0187	0.0167	0.0114	0.0107
sim15	0.00	0	250	0.0442	0.0394	0.0287	0.0237
sim15	0.00	0	1000	0.0204	0.0179	0.0128	0.0116
sim15	0.15	3	250	0.0385	0.0330	0.0241	0.0225
sim15	0.15	3	1000	0.0174	0.0160	0.0119	0.0101
sim15	0.50	3	250	0.0412	0.0354	0.0273	0.0226
sim15	0.50	3	1000	0.0182	0.0164	0.0123	0.0107
sim15	0.85	3	250	0.0415	0.0378	0.0270	0.0236
sim15	0.85	3	1000	0.0198	0.0171	0.0127	0.0111

```
P |>
  mutate(fwer = as.integer(padj_fwer < ALPHA)) |>
  group_by(arm, rho, k, n, B, dataset, item) |>
  summarise(rate = mean(fwer, na.rm = TRUE), .groups = "drop") |>
  group_by(arm, rho, n, B) |>
  summarise(unstable = 100 * mean(rate > 0 & rate < 1), .groups = "drop") |>
  ggplot(aes(factor(B), unstable, colour = rho, group = rho)) +
  geom_line() + geom_point() +
  facet_grid(arm ~ n, labeller = label_both) +
  labs(x = "Bootstrap iterations (B)",
       y = "% of items with a non-unanimous FWER flag",
       colour = "rho") +
  theme_minimal(base_size = 10)
```


